Mandatory deliverable

LAB | Create custom AMI and copy to a different region Lab

Introduction

In this hands-on lab, we will create a custom Amazon Machine Image (AMI) from an EC2 instance and copy it to a different AWS region.

Objectives

By the end of this lab, you will be able to:

1. **Launch an EC2 instance and install required software.**
2. **Create a custom AMI from an existing EC2 instance.**
3. **Copy the AMI to a different AWS region.**
4. **Deploy a new EC2 instance using the copied AMI.**
5. **Verify the installation on the new instance.**
6. **Clean up resources to optimize costs.**

Lab Instructions

1. Launch an EC2 Instance and Install Software

- **Create an EC2 instance** named `master-ec2`:
  - **AMI:** Amazon Linux 2
  - **Instance Type:** `t2.micro`
  - **Key Pair:** Proceed without a key pair.
  - **Network Settings:** Use the default VPC and subnet.
- **Connect to the instance using EC2 Instance Connect.**
- **Install the Apache HTTP Server (`httpd`).**

Ask AI Tutor

```
sudo yum update -y
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd
```

Copy Explain

- **Verify the installation:**

```
systemctl status httpd
```

Copy Explain

2. Create a Custom AMI

- **Select the `master-ec2` instance** in the AWS Management Console.
- **Create an image** (AMI):
  - **Name:** `master-ami`
  - **Description:** Custom AMI with Apache HTTP Server installed.
  - **Confirm and create the image.**

3. Copy the AMI to Another Region

- **Navigate to the AMI section** in the EC2 console.
- **Select the `master-ami` image** and choose **Copy AMI**.
- **Set the destination region** to `US West (Oregon) (us-west-2)`.
- **Initiate the copy process.**

4. Deploy an EC2 Instance from the Copied AMI

- **Switch to `US West (Oregon)` region.**
- **Launch an EC2 instance** named `sec-ec2` using the copied AMI (`master-ami`).
  - **Instance Type:** `t2.micro`
  - **Key Pair:** Proceed without a key pair.
  - **Network Settings:** Use the default VPC and subnet.

5. Verify Installation on the New Instance

- **Connect to `sec-ec2` using EC2 Instance Connect.**
- **Check if Apache HTTP Server is installed and running:**

```
systemctl status httpd
```

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6. Cleanup Resources

- **Terminate EC2 instances (`master-ec2` and `sec-ec2`).**
- **Delete AMIs (`master-ami`).**
- **Remove associated snapshots.**

Completion Criteria

- Custom AMI (`master-ami`) created from `master-ec2` .
- AMI successfully copied to `US West (Oregon)` .
- New EC2 instance (`sec-ec2`) deployed using copied AMI.
- Successful verification of Apache HTTP Server installation.
- All resources cleaned up after the lab.

Submission Guidelines

- Take **screenshots** of each major step.
- Save them into a **document (Google Docs, PDF, etc.)**.
- Upload to **Google Drive** and share the link.
- Ensure **instructor has access** before submission.

🚀 **Great job! You've successfully created and copied a custom AMI across AWS regions!** 🎉

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Lab

URL

Enter the URL here

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